

**Somaiya Vidyavihar University**  
**K J Somaiya School of Engineering**

**Batch:**                      **Roll No.:**  
**Experiment / assignment / tutorial No. 02**  
**Grade: AA / AB / BB / BC / CC / CD /DD**  
**Signature of the Staff In-charge with date**

**Title: Phase Shift Keying**

**Objectives:**

- 1. To implement Phase Shift Keying (PSK) digital modulation technique (transmitter)**
- 2. To write a program for PSK in MATLAB**

**OUTCOME: analyze different band pass digital transmission and reception of signal**

**Draw Circuit Diagram:**

**Procedure:**

- Assemble the circuit as shown in circuit diagram
- Apply digital modulating signal and analog carrier signals with given frequency and amplitude.
- Observe the PSK output on DSO and tabulate the readings.
- Draw the waveforms of modulating, carrier and PSK signal.
- Write a program in MATLAB to observe the PSK waveforms.

**Observation Table:**

| Modulating Signal |           | Carrier signal |           | BPSK Signal |                                   |                                   |
|-------------------|-----------|----------------|-----------|-------------|-----------------------------------|-----------------------------------|
| Amplitude         | Frequency | Amplitude      | Frequency | Amplitude   | Phase shift when<br>Logic 0 input | Phase shift when<br>Logic 1 input |
|                   |           |                |           |             |                                   |                                   |

**Discussion/Conclusion:**

**Signature of faculty in-charge**